

Backlog Buddy

Demonstration question paper generated from available previous papers - for practice only

B.Tech. I Semester Examination, 2024

Regulation: R20 | University: JNTUK

Data Structures (21CS32)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Introduction & Linear Structures]

- a) Compare array and linked list representations of a list with suitable examples.
- b) Compare array and linked list representations of a list with suitable examples.

UNIT - 2 [Stacks, Queues & Recursion]

- a) Write an algorithm to convert an infix expression into postfix form using a stack.
- b) Describe the implementation of a circular queue and its advantages over a linear queue.

UNIT - 3 [Trees & Balanced Search Trees]

- a) Construct a binary search tree for the given data and perform insertion and deletion.
- b) Explain AVL tree rotations with examples.

UNIT - 4 [Graphs & Graph Algorithms]

- a) Explain DFS and BFS traversals of a graph with examples.
- b) Explain Dijkstra algorithm for single source shortest path with an example.

UNIT - 5 [Searching, Sorting & Hashing]

- a) Write the quick sort algorithm and analyse its time complexity.
- b) Explain merge sort with a recursion tree and derive its complexity.

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